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COMPUTER NETWORKS

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Ex.No:16	**NMAP TO DISCOVER LIVE HOSTS USING ARP, ICMP, TCP AND UDP PING SCANS IN TRYHACKME**
DATE:	

1. Aim

To discover live hosts in a network using Nmap by performing ARP scan, ICMP scan, TCP ping scan, and UDP ping scan through a free TryHackMe learning environment.

2. Software Requirements

- Computer or laptop with an internet connection
- Web browser
- Free TryHackMe account
- A free-access lab room with an authorized target
- Nmap terminal in the AttackBox or an authorized lab machine

3. Procedure

Step 1: Create a free TryHackMe account

Open https://tryhackme.com in a web browser. Select Sign Up, register using an available sign-in method, and log in to the account.

Step 2: Select a free room

Visit **https://tryhackme.com/module/nmap** and look for a room or activity related to Nmap and host discovery. Confirm that it is available under the free plan. If the relevant activity requires a subscription, choose another free room that provides a suitable authorized lab target. The learning-path catalogue is available at https://tryhackme.com/paths.

Step 3: Start the lab environment

Open the selected room, read its instructions, and start the target machine. Start the browser-based AttackBox if it is included in the available plan. Note the target IP address shown by the room and open the terminal. If using a local terminal, ensure it can reach the authorized lab target.

Step 4: Verify Nmap installation

Enter the following command in the terminal:

nmap --version

Step 5: Perform ARP scan
Command:

sudo nmap -PR -sn TARGET_IP

Explanation:

- -PR: Uses ARP requests for host discovery.
- -sn: Discovers hosts without performing a port scan.
- TARGET_IP: Replace this placeholder with the IP address of the authorized lab target.

Observation: Record whether Nmap identifies the target as up. ARP discovery generally applies to the same local network segment.

Step 6: Perform ICMP scan
Command:

sudo nmap -PE -sn TARGET_IP

Explanation: -PE sends ICMP Echo Request probes. The -sn option performs host discovery without port scanning.

Observation: Record whether Nmap reports “Host is up”. A firewall may block ICMP requests even when a host is active.

Step 7: Perform TCP ping scan
Command:

sudo nmap -PS80,443 -sn TARGET_IP

Explanation: -PS sends TCP SYN probes to ports 80 and 443. The -sn option disables port scanning.

Observation: Record whether Nmap detects the target as reachable. The result depends on target configuration and firewall rules.

Step 8: Perform UDP ping scan
Command:

sudo nmap -PU53,123 -sn TARGET_IP

Explanation: -PU sends UDP host-discovery probes to ports 53 and 123. The -sn option performs host discovery without port scanning.

Observation: Record the actual output. UDP probes may receive no response even when the target is active.

4. Observation Table

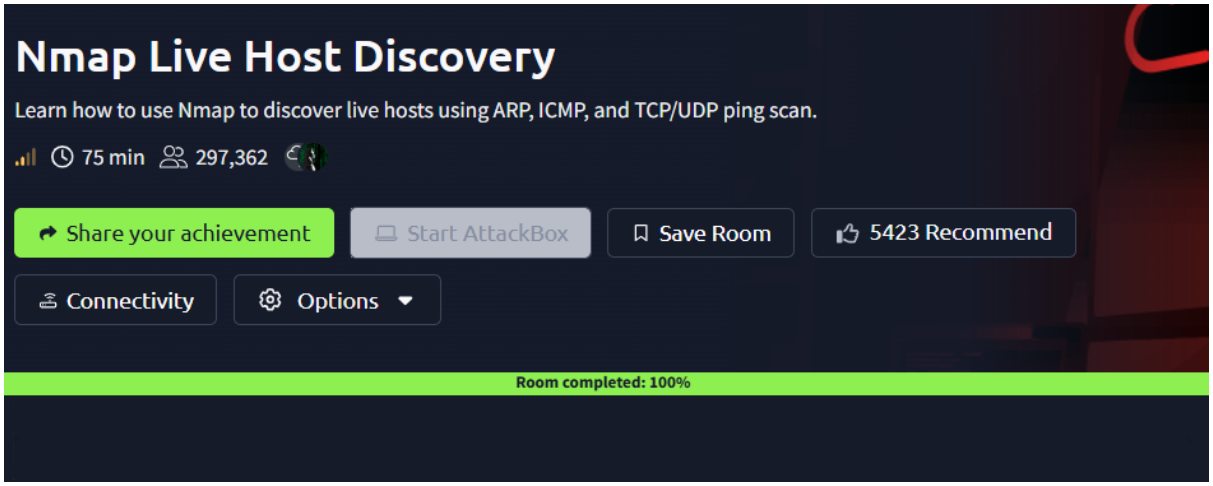
S. No.	Scan Technique	Nmap Option	Observed Result
1	ARP scan	-PR	_____
2	ICMP scan	-PE	_____
3	TCP SYN ping	-PS	_____

4	UDP ping	-PU	_____

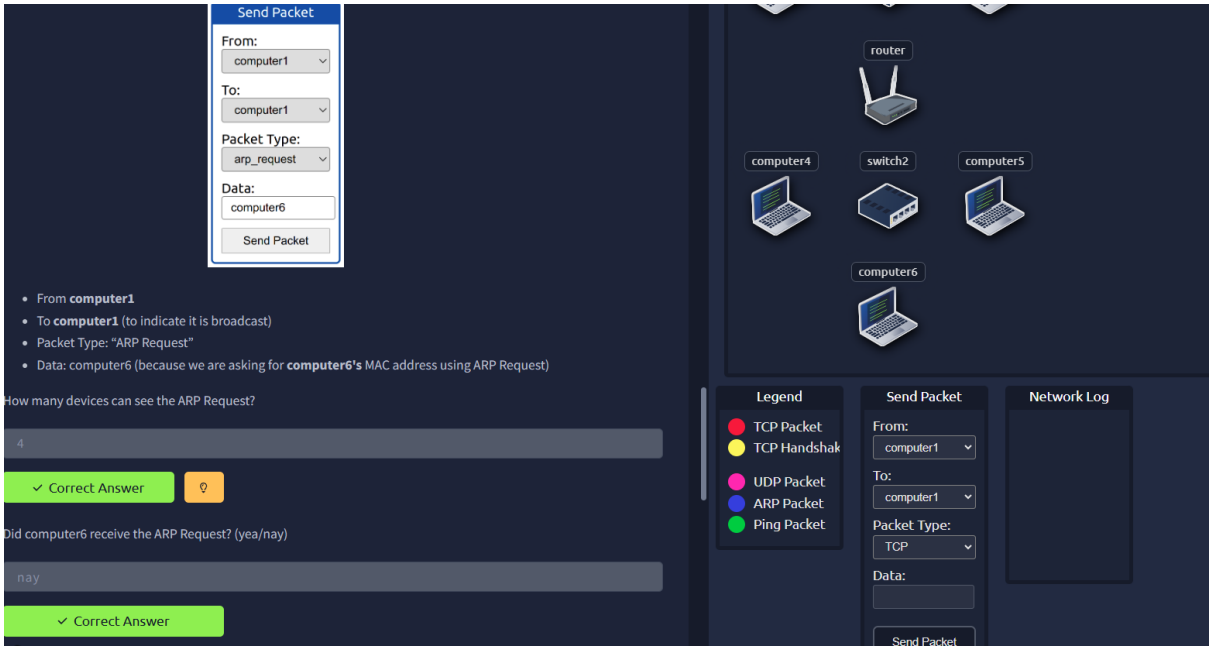
Expected observation table

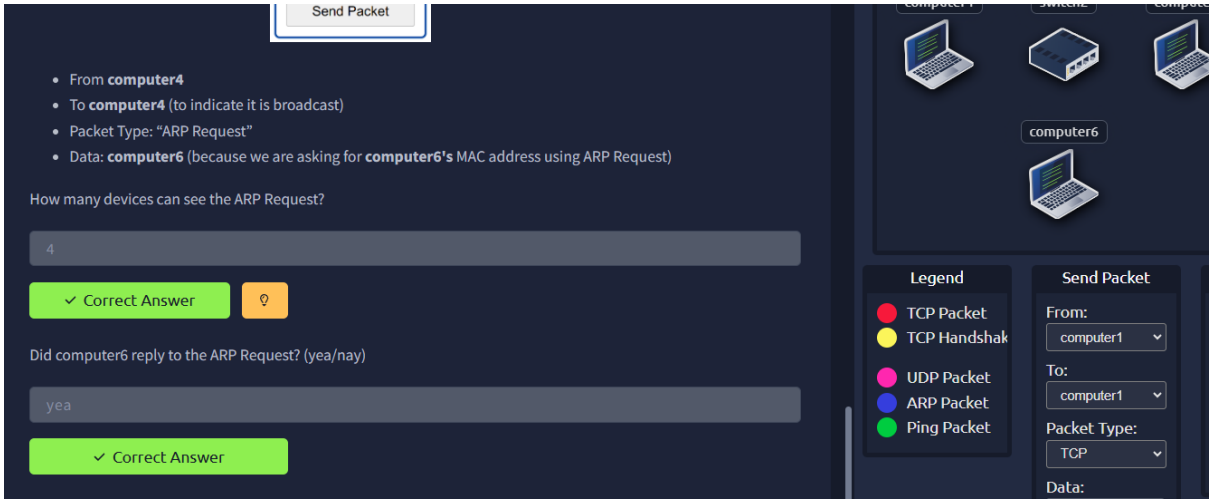
S. No.	Scan technique	Command option	Expected observation
1	ARP scan	`-PR`	Host detected on the same local network
2	ICMP scan	`-PE`	Host detected if ICMP response is received
3	TCP SYN ping	`-PS`	Host detected if a suitable TCP response is received
4	UDP ping	`-PU`	Host detected if a suitable response is received

OUTPUT:






Send Packet
From:
computer1
To:
computer1
Packet Type:
arp_request
Data:
computer6
Send Packet
router
computer4
switch2
computer5
computer6
From computer1
To computer1 (to indicate it is broadcast)
Packet Type: "ARP Request"
Data: computer6 (because we are asking for computer6's MAC address using ARP Request)
How many devices can see the ARP Request?
4
Correct Answer
Did computer6 receive the ARP Request? (yea/nay)
nay
Correct Answer
Legend
TCP Packet
TCP Handshak
UDP Packet
ARP Packet
Ping Packet
Send Packet
From:
computer1
To:
computer1
Packet Type:
TCP
Data:
Send Packet
Network Log


Send Packet
From computer4
To computer4 (to indicate it is broadcast)
Packet Type: "ARP Request"
Data: computer6 (because we are asking for computer6's MAC address using ARP Request)
How many devices can see the ARP Request?
4
Correct Answer
Did computer6 reply to the ARP Request? (yea/nay)
yea
Correct Answer
computer6
Legend
TCP Packet
TCP Handshak
UDP Packet
ARP Packet
Ping Packet
Send Packet
From:
computer1
To:
computer1
Packet Type:
TCP
Data:

Send a packet with the following:
From **computer1**
To **computer3**
Packet Type: "**Ping Request**"
What type of packet did **computer1** send before the ping?

ARP Request

✓ Correct Answer

What type of packet did **computer1** receive before it was able to send the ping?

ARP Response

✓ Correct Answer

How many computers responded to the ping request?

1

✓ Correct Answer

Send a packet with the following:
From **computer2**
To **computer5**
Packet Type: "**Ping Request**"
What is the name of the first device that responded to the first ARP Request?

router

✓ Correct Answer

What is the name of the first device that responded to the second ARP Request?

computer5

✓ Correct Answer

Send another Ping Request. Did it require new ARP Requests? (yea/nay)

nay

✓ Correct Answer

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8

✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400

✓ Correct Answer

Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

1

✓ Correct Answer

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

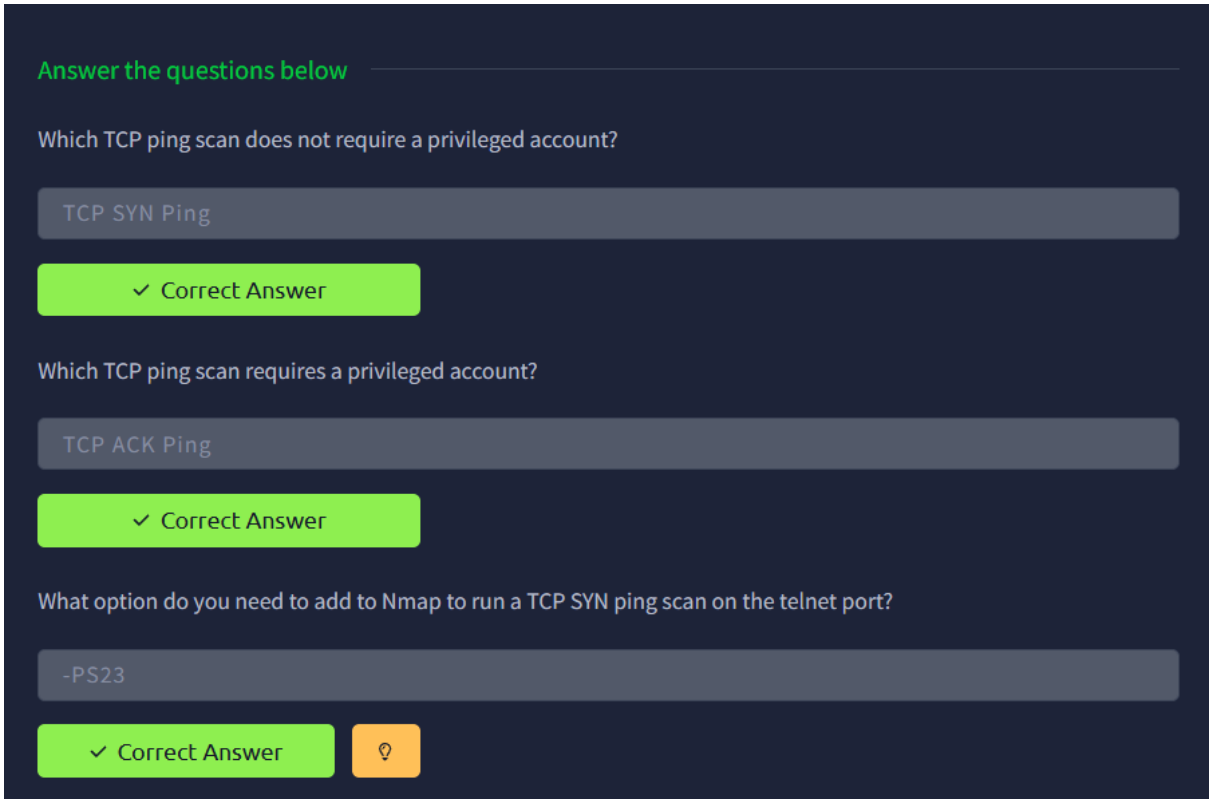
-PM

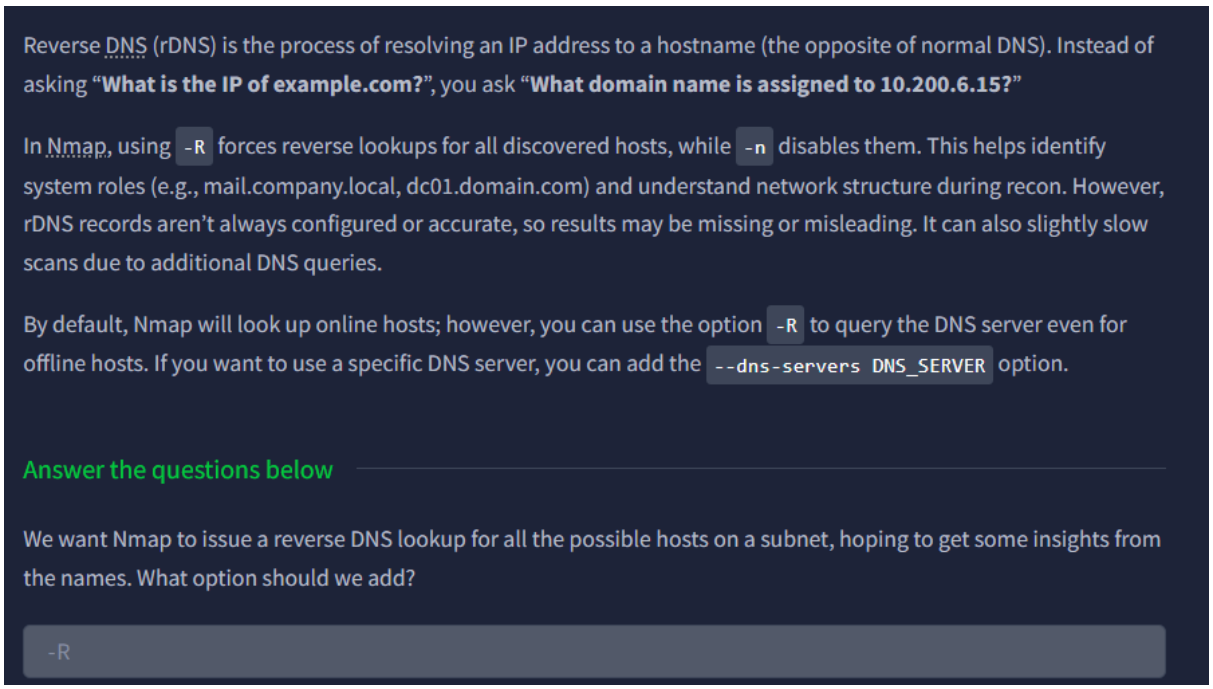
✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE

✓ Correct Answer





Result:

Live host discovery was studied using Nmap in the TryHackMe learning environment. ARP, ICMP, TCP, and UDP probes were used to examine host reachability, and the observations were recorded.